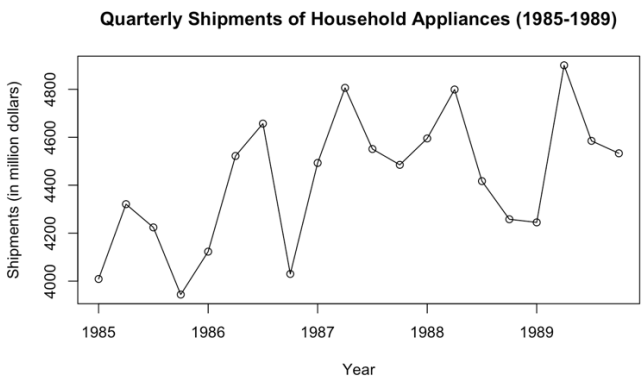
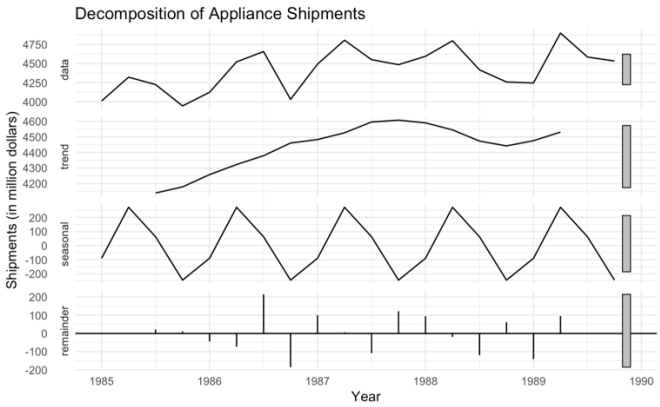
Business Data Mining Homework Session 6

18.6 Shipments of Household Appliances. The time plot in Figure 18.9 shows the series of quarterly shipments (in million dollars) of US household appliances between 1985-1989 (data are available in ApplianceShipments.csv, data courtesy of Ken Black).

a. Which of the following methods would be suitable for forecasting this series if applied to the raw data?
- Moving average
- Simple exponential smoothing
- Double exponential smoothing
- Holt-Winter's exponential smoothing





Firstly, let's have a look of the raw data. The chart on the right shows 4 components of the shipments data: level, trend, additive seasonality and error. The error component has significant influence on the stability of the seasonality.

Moving average

This method is a straightforward and easily interpretable forecasting technique. It could smooth out data fluctuations, averaging out the peaks and troughs to reveal underlying patterns. However, it is suitable for data without obvious trend or seasonality. According to the chart, we know that shipment data has a clear trend and strong seasonality; the moving average method may fail to capture these essential characteristics. I think it may not be suitable for shipment data.

Simple exponential smoothing

It is a reliable forecasting method when the data fluctuates around a stable value. However, this method could not capture trend and seasonality. As discussed before, the shipment data shows both a well-defined trend and strong seasonal components and is not fluctuating around a certain value. Therefore, I think this method is not suitable for forecasting shipment data.

Double exponential smoothing

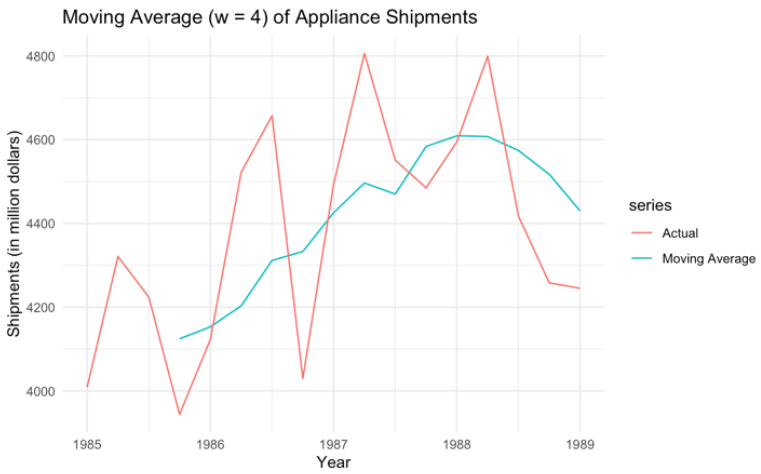
Double exponential method is an extension of the simple exponential smoothing method that explicitly accounts for the trend component in a time series. It forecasts a project as a straight line, capturing a

persistent growth or decline. However, double exponential method could not capture the seasonal component. Therefore, I think it is not suitable for forecasting the shipment data.

Holt-Winter's exponential smoothing
This method captures the level, trend, and seasonality of a time series. It is straightforward to implement and easy to understand the data. The method provides highly accurate forecasts for short-term horizons, especially for data with obvious trend and seasonality. However, the accuracy may diminish when predicting long-term data due to the cumulative uncertainty in trend and seasonality. Therefore, I think it is suitable for forecasting shipment data.

b. Apply a moving average with window span w = 4 to the data. Use all but the last year as the training set. Create a time plot of the moving average series.

i. What does the MA(4) chart reveal?



The application of a 4-period moving average effectively smooths the volatile fluctuations in the original shipment data by averaging out the peaks and troughs. While the actual data clearly reveals a strong seasonal pattern, the moving average line acts like a filter, suppressing the seasonality component to reveal the underlying trend.

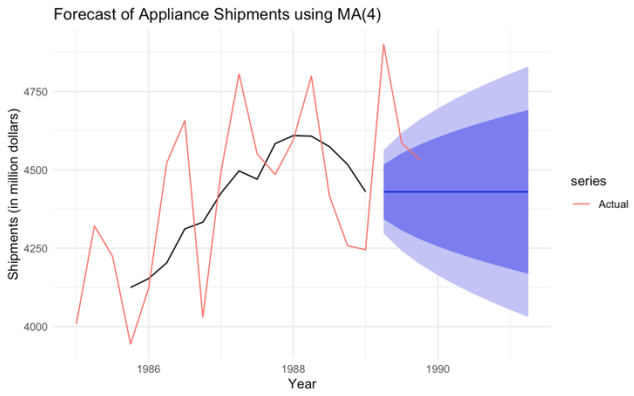
ii. Use the MA(4) model to forecast appliance sales in Q1-1990.

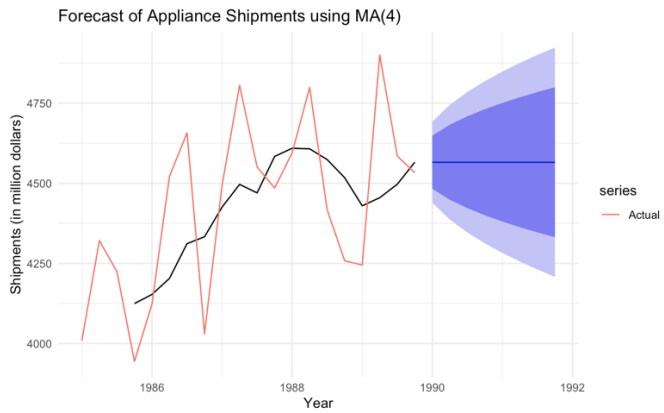
Forecast for Q1–1990 is 4,403 million dollars.

iii. Use the MA(4) model to forecast appliance sales in Q1-1991.

Forecast for Q1–1991 is 4,261 million dollars.

iv. Is the forecast for Q1-1990 most likely to under-estimate, over-estimate or accurately estimate the actual sales on Q1-1990? Explain.





I think the forecast for Q1 1990 is most likely to be underestimated. This is because the moving average method lags behind the actual data. Given the clear increasing trend in the shipment data and the fact that Q1 typically experiences higher seasonal demand. The moving average method is based on the average of past data, which might fail to capture the immediate increase in sales, resulting in a forecast that is lower than the actual value.

v. Management feels most comfortable with moving averages. The analyst therefore plans to use this method for forecasting future quarters. What else should be considered before using the MA(4) to forecast future quarterly shipments of household appliances?

While the moving average method is easy to understand, it is crucial to acknowledge its shortcomings for this shipment data before proceeding. This method could not capture the clear trend and strong seasonality, lagging behind the actual value and producing inaccurate forecasts. To align management's comfort with analytical rigor, I would recommend:

- Using different methods simultaneously, such as Arima or Holt-Winter's exponential smoothing, which helps to compare different forecasting results.
- Strictly limit this method to short-term forecasts because errors accumulate quickly.
- Test different window lengths to find out the most suitable window span and to minimize the recent error.

c. We now focus on forecasting beyond 1989. In the following, continue to use all but the last year as the training set, and the last four quarters as the validation set. First, fit a regression model to sales with a linear trend and quarterly seasonality to the training data. Next, apply Arima's exponential smoothing (with the default smoothing values) to the training data. Choose an adequate "season length."

i. Compute the MAPE for the validation data using the regression model.

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1	Theil's U
Training set	0.00	151.56	128.11	-0.12	2.91%	0.58	0.28	NA
Validation set	-18.97	99.01	81.73	-0.41	1.78%	0.37	-0.62	0.54

According to the results in the table, the regression model demonstrates a strong predictive accuracy. The MAPE of training set is 2.91% and the MAPE of validation data is 1.78%, which is lower than the figure of training set MAPE. Usually, the MAPE lower than 10% is enough to show the model is good.

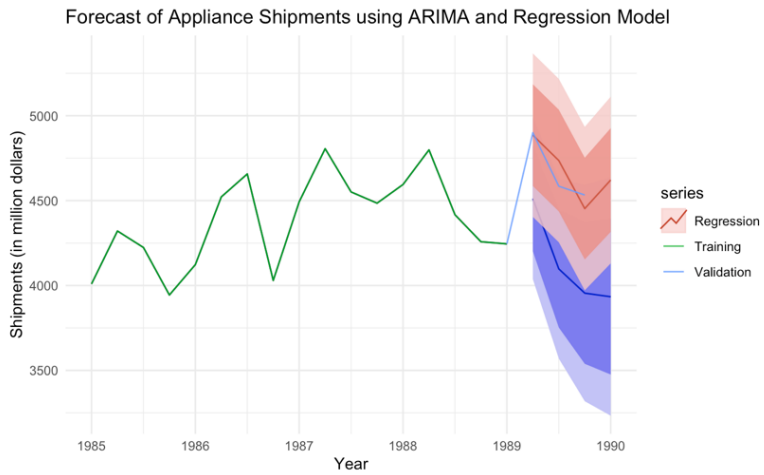
The RMSE of training data is 151.56 and RMSE of validation data is 99.01, which shows the model has a good performance on validation data, generalize well on unseen data and is not overfit.

ii. Compute the MAPE for the validation data using Arima model.

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1	Theil's U
Training set	-38.55	194.62	142.50	-0.91	3.19%	0.65	-0.26	NA
Validation set	484.50	490.65	484.50	10.43	10.43%	2.20	-0.00	2.46

The ARIMA model exhibits a clear sign of overfitting. The MAPE of training set is 3.19%, but the MAPE of validation set is 10.43%, which is much higher than the training set MAPE. This issue is further confirmed by the high positive value of validation set ME, which reveals a systemic forecasting bias. Therefore, this Arima model is not a good choice for shipment data.

iii. Which model would you prefer to use for forecasting Q1-1990? Give three reasons.



According to the above analysis, I would recommend the regression model for 3 reasons. Firstly, it has better performance, especially its validation set MAPE is much lower than Arima's. Secondly, the Arima model has a clear overfitting problem, while the regression model does not. Thirdly, the regression model's stability and accuracy on the validation set make it more reliable and trustworthy.

iv. If we optimize the smoothing parameters in the Arima method, is it likely to get values that are close to zero? Why or why not?

No, given the clear trend and strong seasonality in the shipment data, the optimized parameters are very unlikely to be near zero. A near-zero trend parameter would incorrectly model the data with no growth. A near-zero seasonality parameter would ignore the seasonal fluctuations. Therefore, there is no likelihood of getting values close to 0. Since the optimization goal is to create an accurate model, it must choose parameters to correctly reflect the true data characteristics.